1 Stationarity tests

Table 1: ADF (H0: unit root) and KPSS (H0: stationary). KPSS 5% critical value = 0.463.

Series	ADF p	ADF	KPSS stat	KPSS	n
TFR (raw)	0.120	Non-stationary	1.071	Non-stationary	53
diff(TFR), d = 1	0.083	Non-stationary	0.841	Non-stationary	52
diff(TFR), d = 2	0.010	Stationary	0.043	Stationary	51
log(TFR)	0.603	Non-stationary	1.201	Non-stationary	53
diff(log TFR), d = 1	0.031	Stationary	0.505	Non-stationary	52
TLB (raw)	0.356	Non-stationary	0.624	Non-stationary	53
diff(TLB), d = 1	0.039	Stationary	0.213	Stationary	52

2 Correlograms of the differenced series

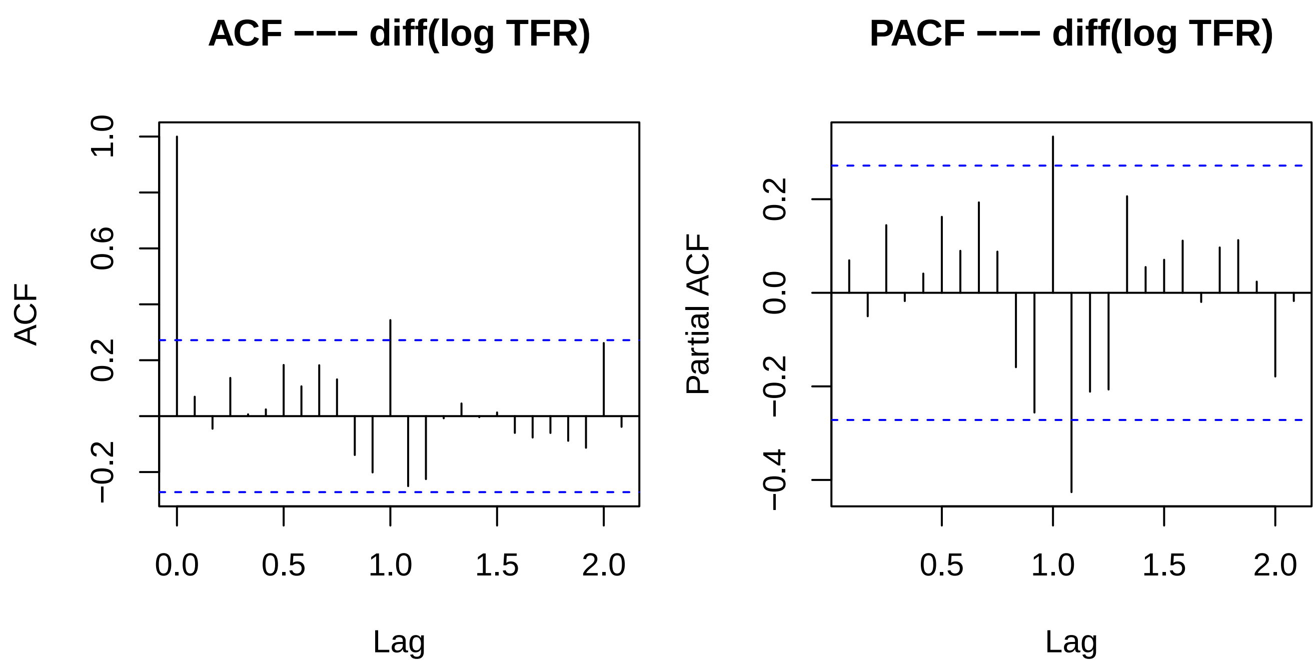


Figure 1: ACF and PACF of diff(log TFR).

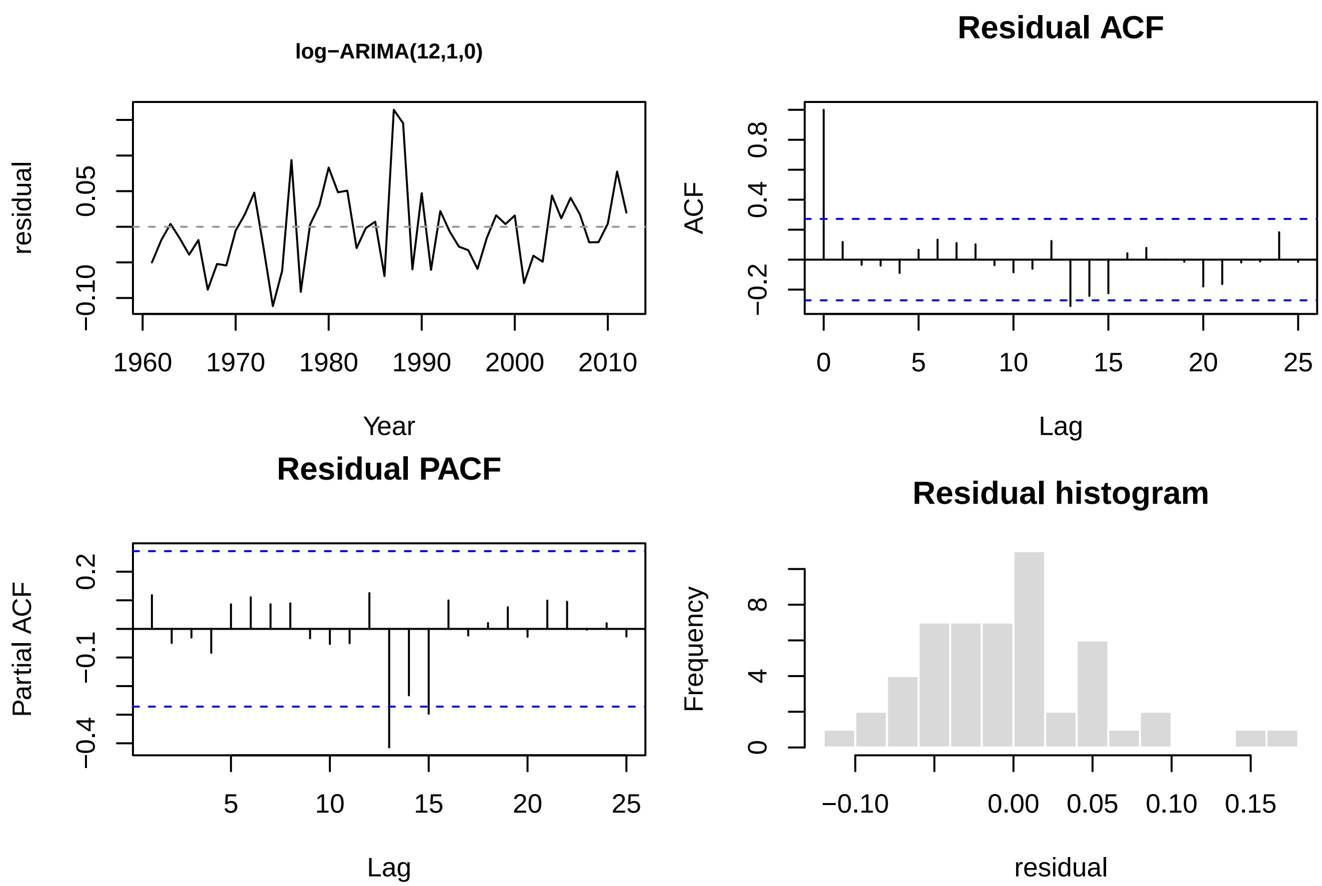
3 High-order ARIMA candidates

Table 2: High-order ARIMA fitted on diff(log TFR) (TFR) and diff(TLB) (TLB).

Model	npar	AIC	BIC	LB Q*	LB df	LB p-value
log-ARIMA(12,1,0)	12	-123.91	-98.55	21.19	3	0.0001
log-ARIMA(13,1,0)	13	-131.80	-104.48	15.48	3	0.0014
ARIMA(11,1,0)	11	985.77	1009.19	23.14	3	0.0000
ARIMA(12,1,0)	12	982.23	1007.60	18.32	3	0.0004

Model	npar	AIC	BIC	LB Q*	LB df	LB p-value
ARIMA(13,1,0)	13	976.61	1003.93	12.94	3	0.0048

3.1 Residual plots — ARIMA candidates



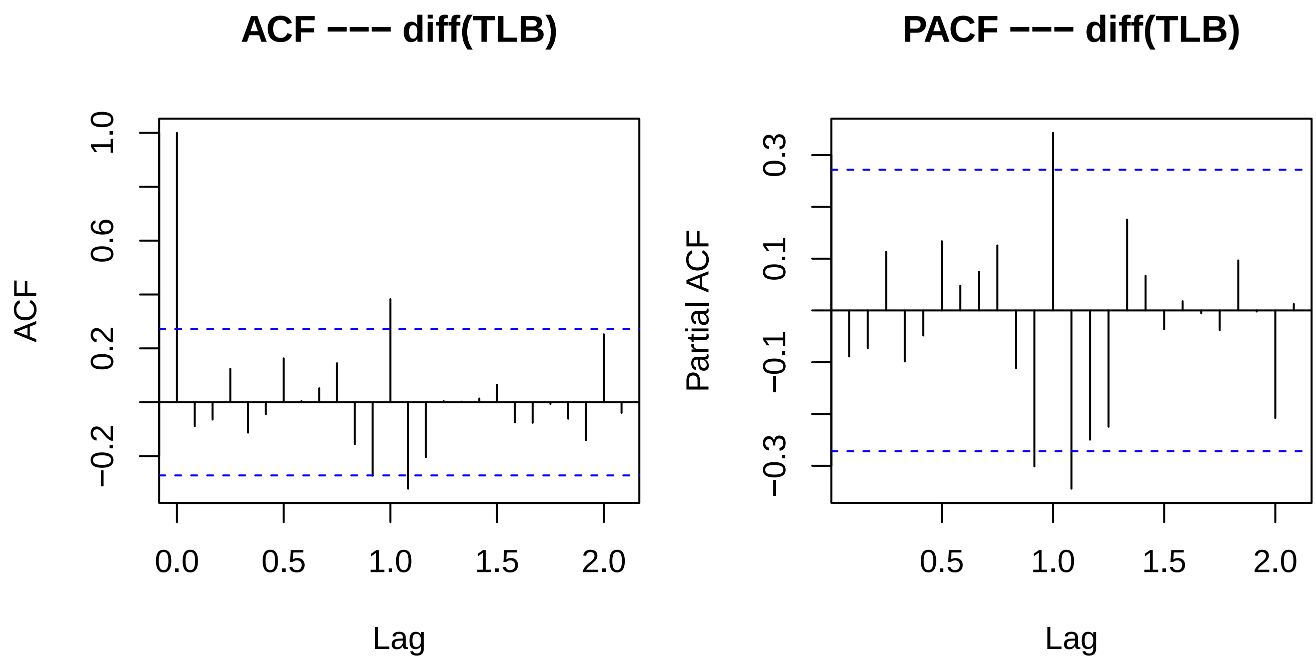
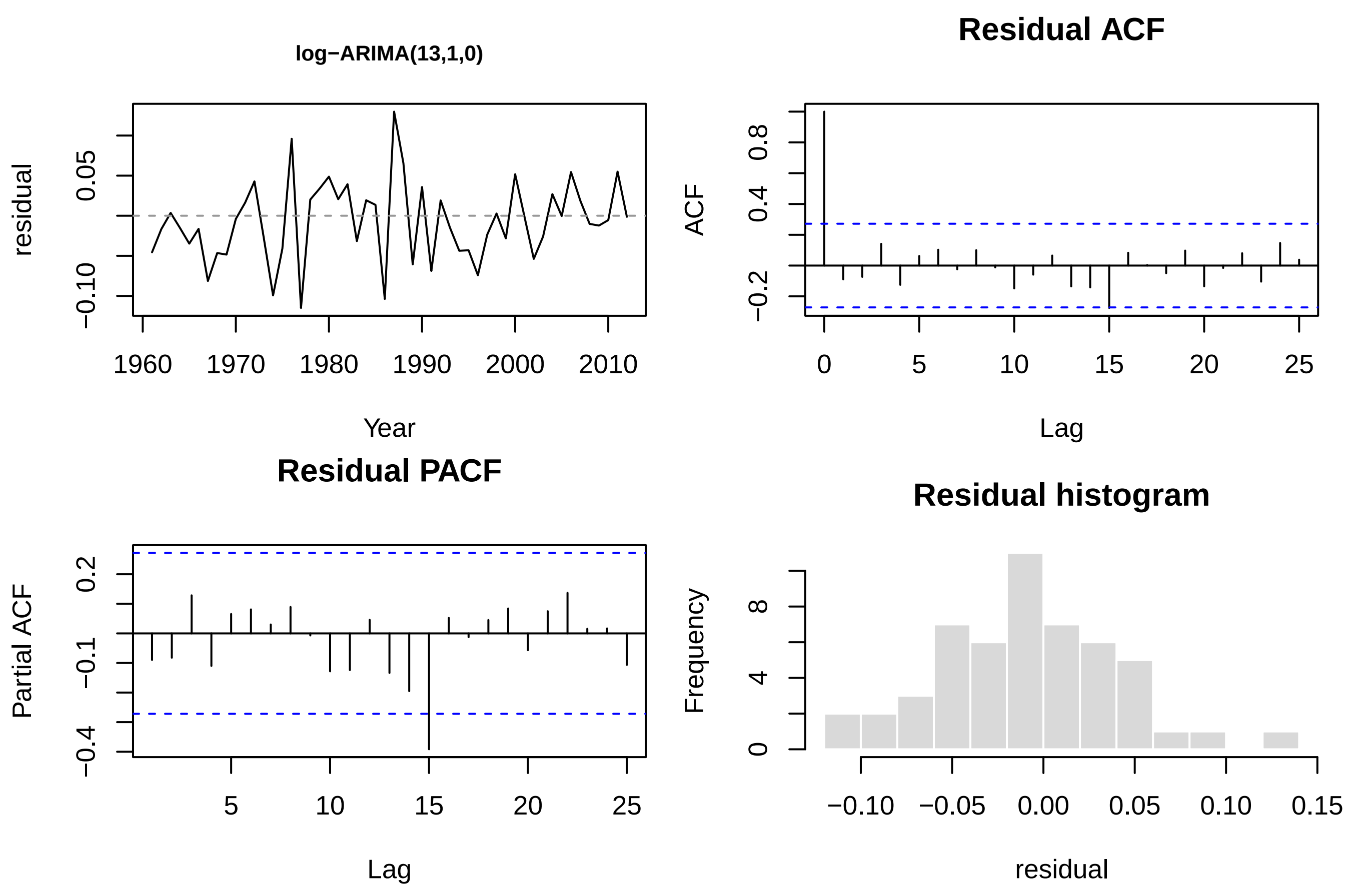
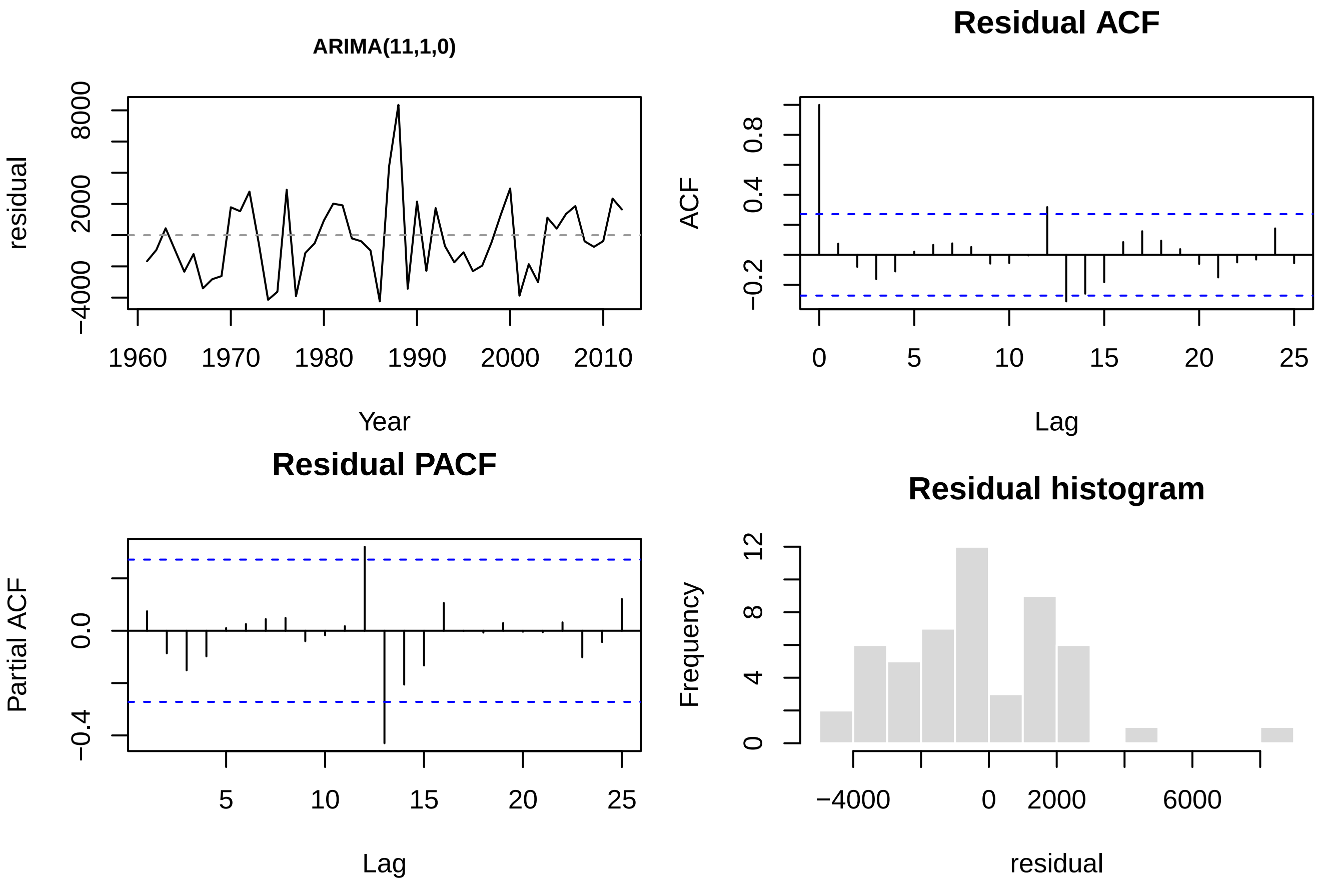
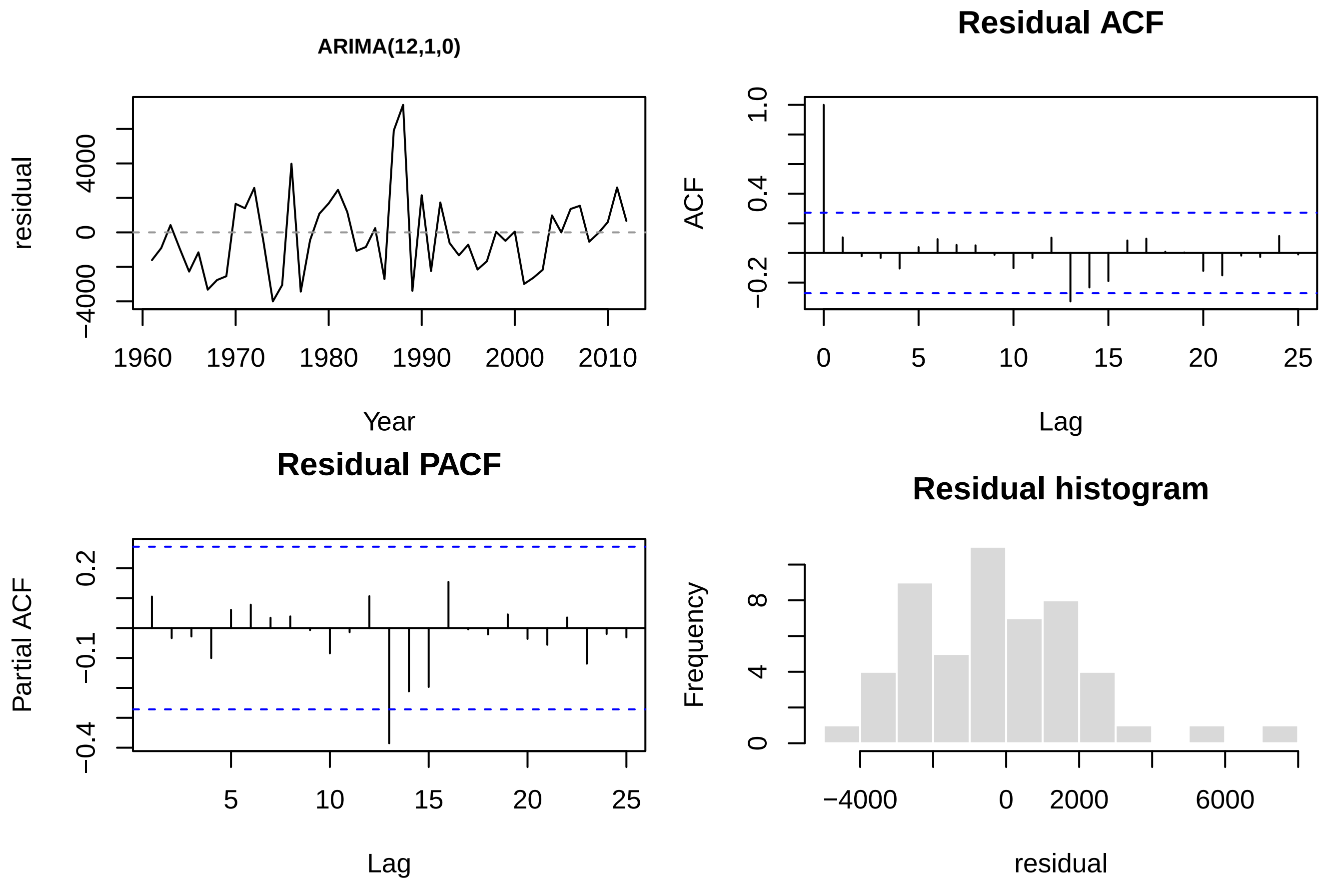
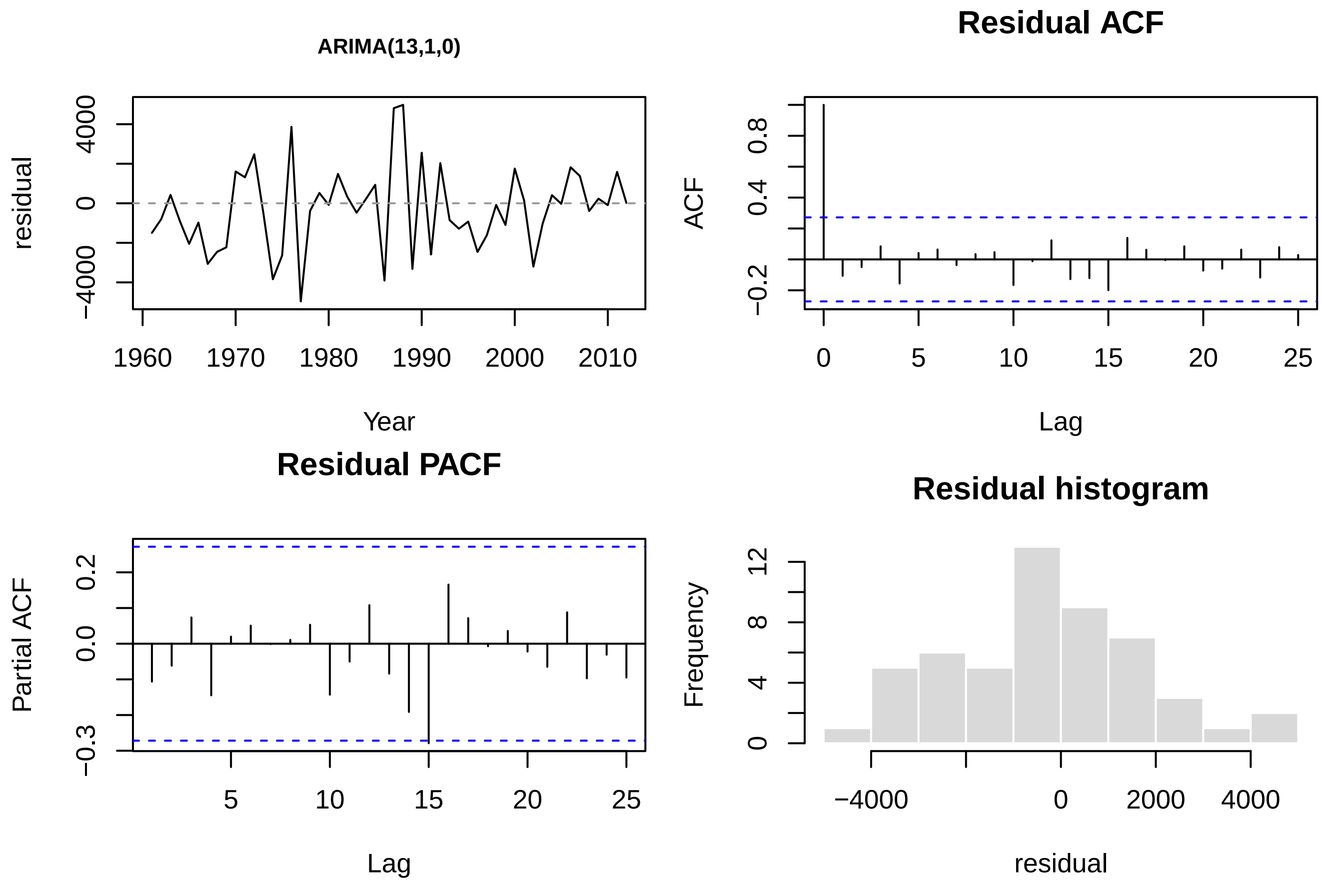


Figure 2: ACF and PACF of diff(TLB).




ARIMA(11,1,0)
residual
−4000
2000
8000
1960
1970
1980
1990
2000
2010
Year
Residual ACF
ACF
−0.2
0.4
0.8
0
5
10
15
20
25
Lag
Residual PACF
Partial ACF
−0.4
0.0
5
10
15
20
25
Lag
Residual histogram
Frequency
0
4
8
12
−4000
0
2000
6000
residual


ARIMA(12,1,0)
residual
−4000
0
4000
1960
1970
1980
1990
2000
2010
Year
Residual ACF
ACF
−0.2
0.4
1.0
0
5
10
15
20
25
Lag
Residual PACF
Partial ACF
−0.4
−0.1
0.2
5
10
15
20
25
Lag
Residual histogram
Frequency
0
4
8
−4000
0
2000
6000
residual



4 Adding the seasonal operator

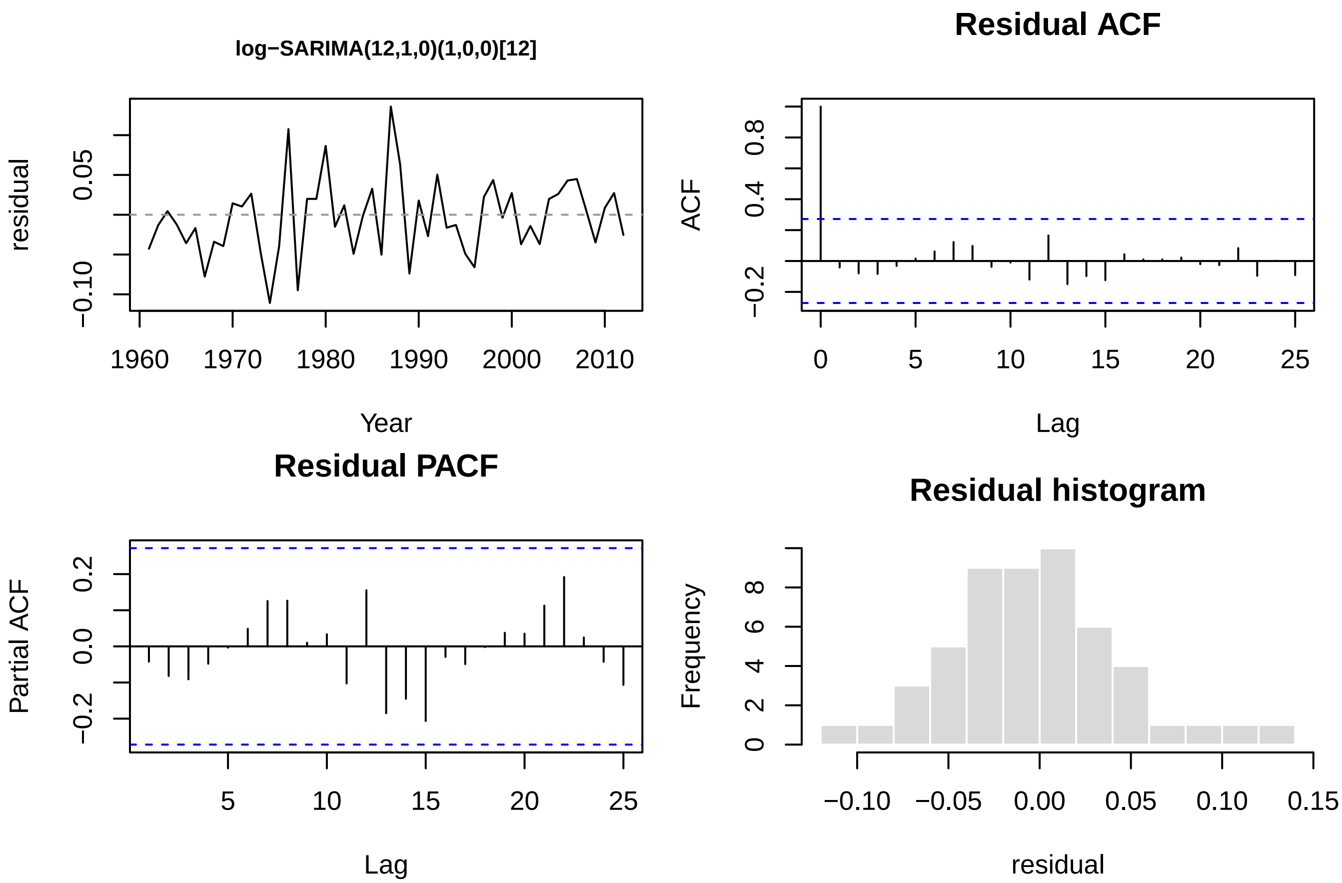
Because the ARIMA residual correlograms still show significant autocorrelation at the seasonal lag — period 12, the 12-year Chinese-zodiac Dragon/Tiger fertility cycle — we extend each model with a **seasonal AR(1) term at lag 12**, giving SARIMA(p, 1, 0)(1, 0, 0)[12].

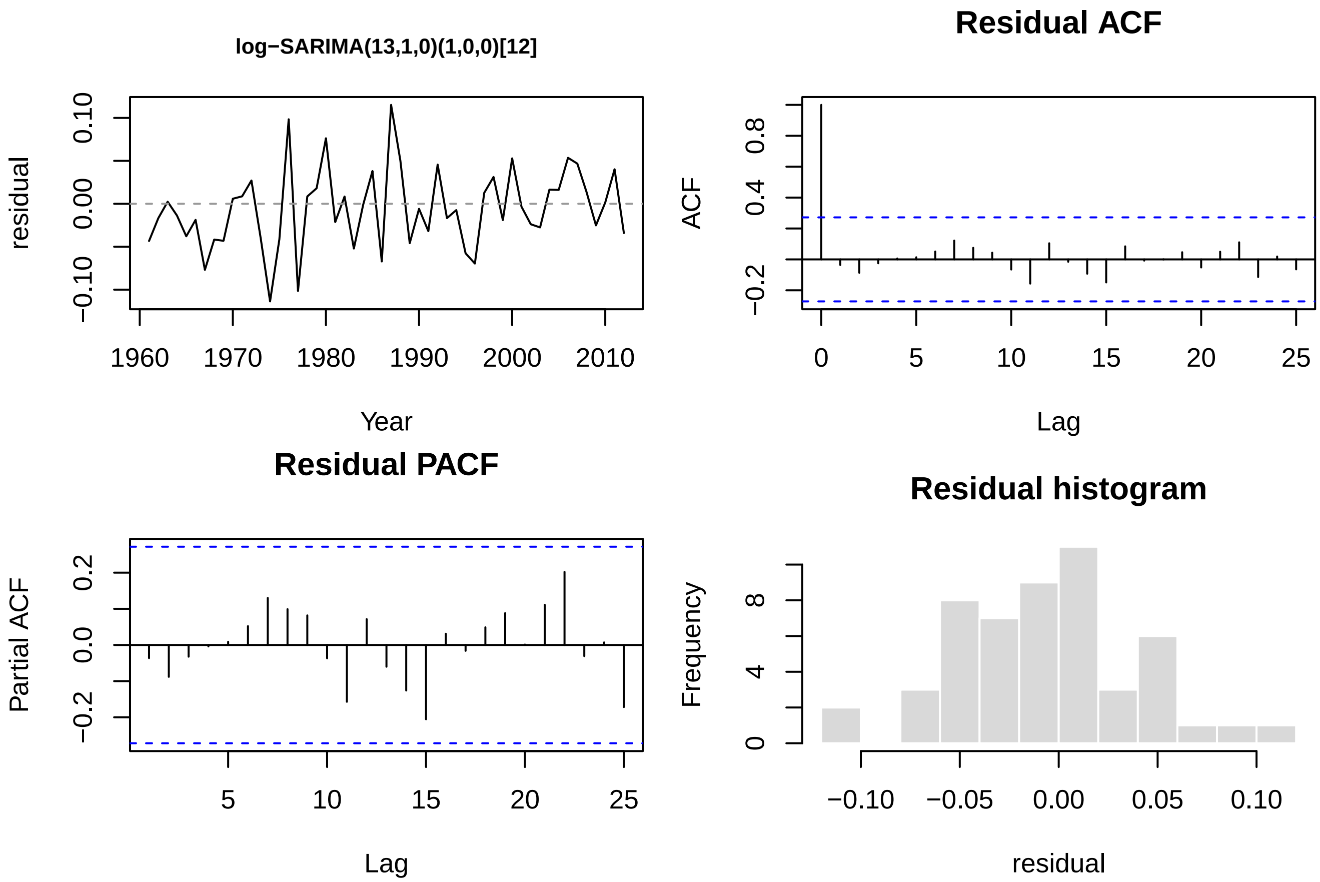
5 SARIMA candidates

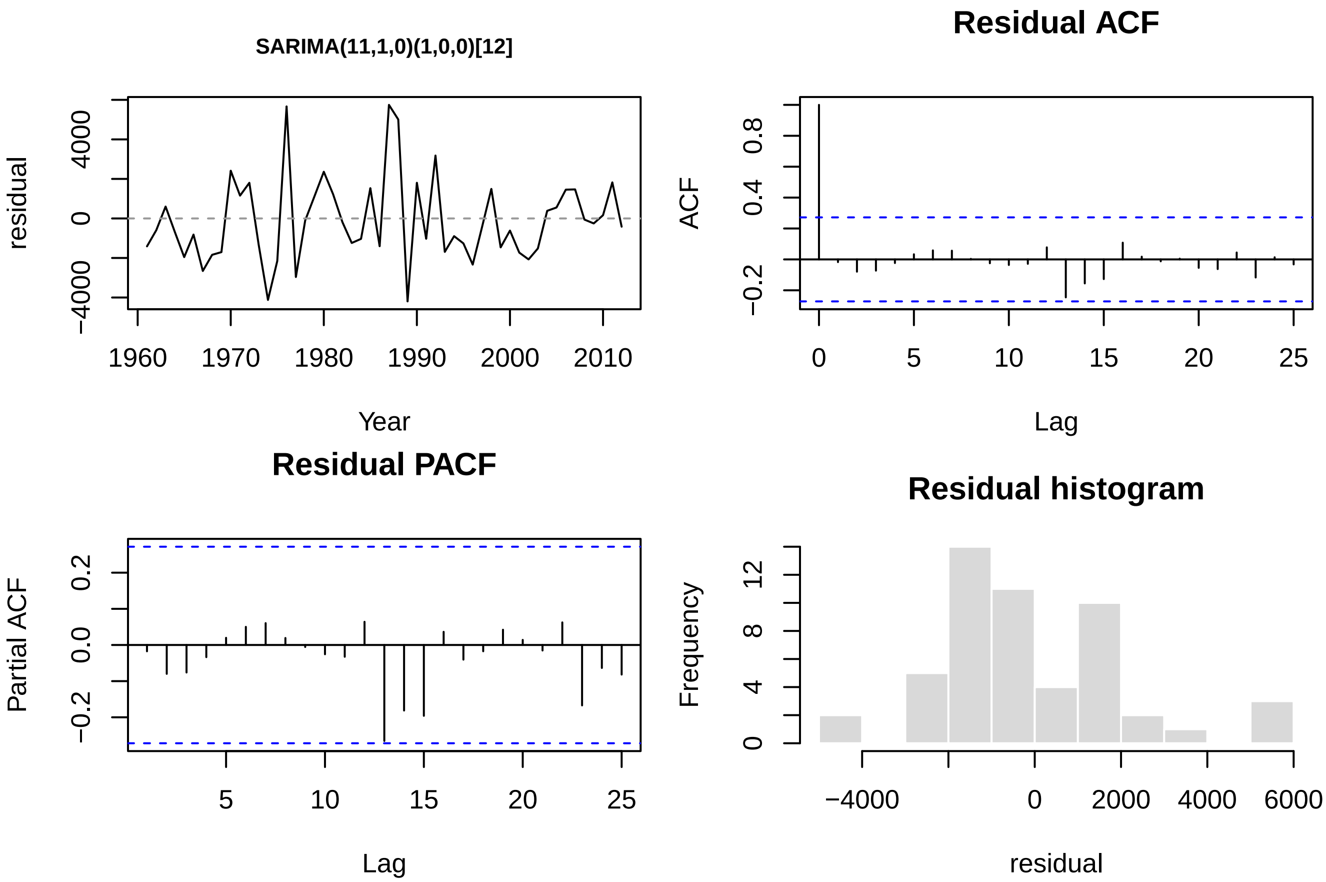
Table 3: SARIMA with a seasonal AR(1) at period 12, fitted on the differenced series. LB = Ljung-Box residual test.

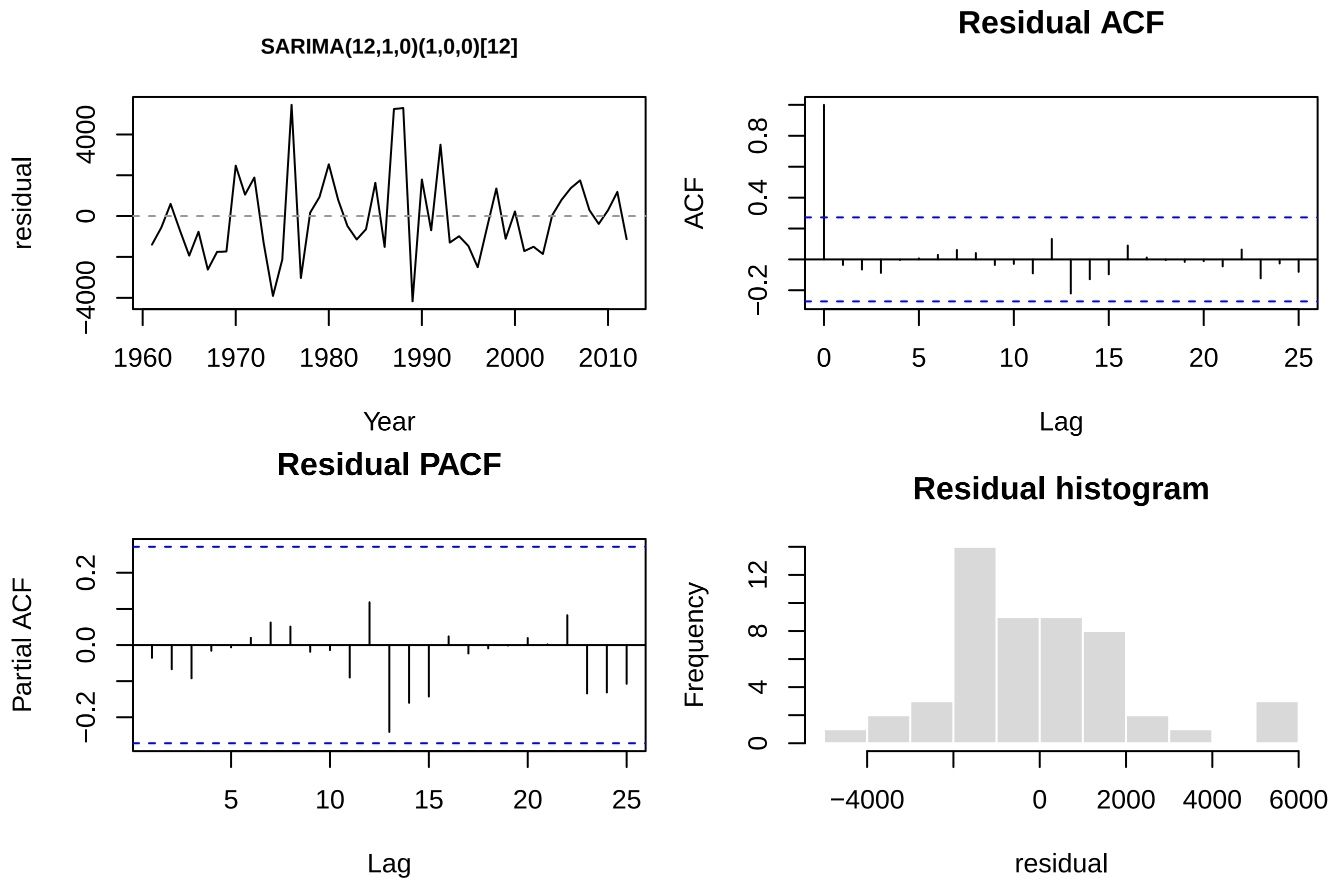
Model	npar	AIC	BIC	LB Q*	LB df	LB p-value
log-SARIMA(12,1,0)(1,0,0)[12]	13	-133.06	-105.75	9.37	3	0.0248
log-SARIMA(13,1,0)(1,0,0)[12]	14	-133.69	-104.42	7.76	3	0.0513
SARIMA(11,1,0)(1,0,0)[12]	12	976.26	1001.62	9.16	3	0.0272
SARIMA(12,1,0)(1,0,0)[12]	13	977.13	1004.44	9.15	3	0.0274
SARIMA(13,1,0)(1,0,0)[12]	14	975.66	1004.93	7.70	3	0.0527

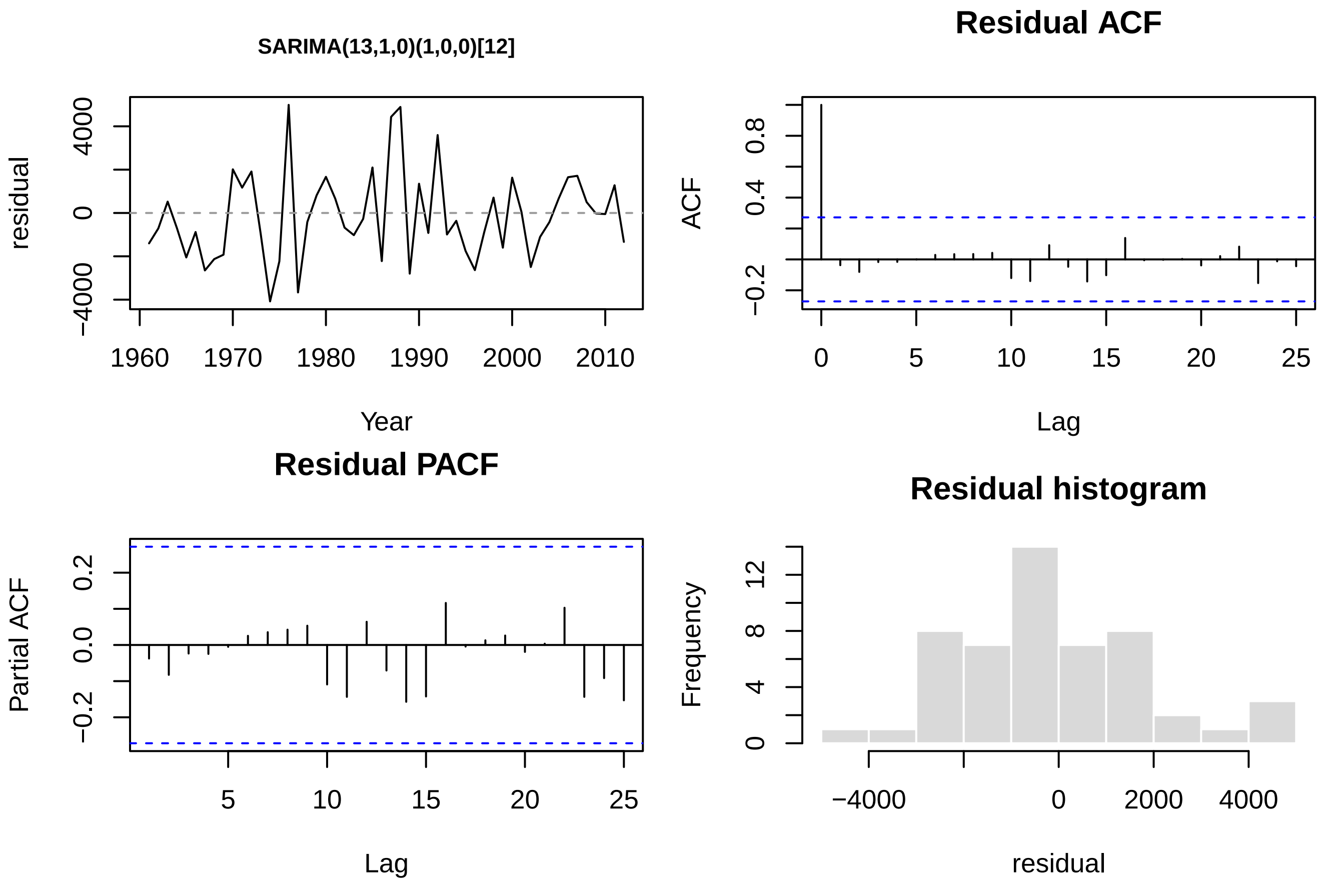
5.1 Residual plots — SARIMA candidates




log−SARIMA(13,1,0)(1,0,0)[12]
residual
−0.10
0.00
0.10
1960
1970
1980
1990
2000
2010
Year
Residual ACF
ACF
−0.2
0.4
0.8
0
5
10
15
20
25
Lag
Residual PACF
Partial ACF
−0.2
0.0
0.2
5
10
15
20
25
Lag
Residual histogram
Frequency
0
4
8
−0.10
−0.05
0.00
0.05
0.10
residual


SARIMA(11,1,0)(1,0,0)[12]
residual
−4000
0
4000
1960
1970
1980
1990
2000
2010
Year
Residual ACF
ACF
−0.2
0.4
0.8
0
5
10
15
20
25
Lag
Residual PACF
Partial ACF
−0.2
0.0
0.2
5
10
15
20
25
Lag
Residual histogram
Frequency
0
4
8
12
−4000
0
2000
4000
6000
residual


SARIMA(12,1,0)(1,0,0)[12]
residual
Year
Residual ACF
ACF
Lag
Residual PACF
Partial ACF
Lag
Residual histogram
Frequency
residual



6 Forecast vs. actual — best models

Our best models — selected for white-noise residuals together with the lowest out-of-sample error — add a **seasonal difference** at lag 12 to the seasonal-AR models above: **log-SARIMA(12,1,0)(0,1,0)[12]** for TFR and **SARIMA(13,1,0)(1,1,1)[12]** for TLB. The plots below compare the 2013–2024 point forecast (red, dashed) against the realised values (blue), with 80% and 95% prediction intervals (the model is fitted on the differenced series and the forecast integrated back to the level).

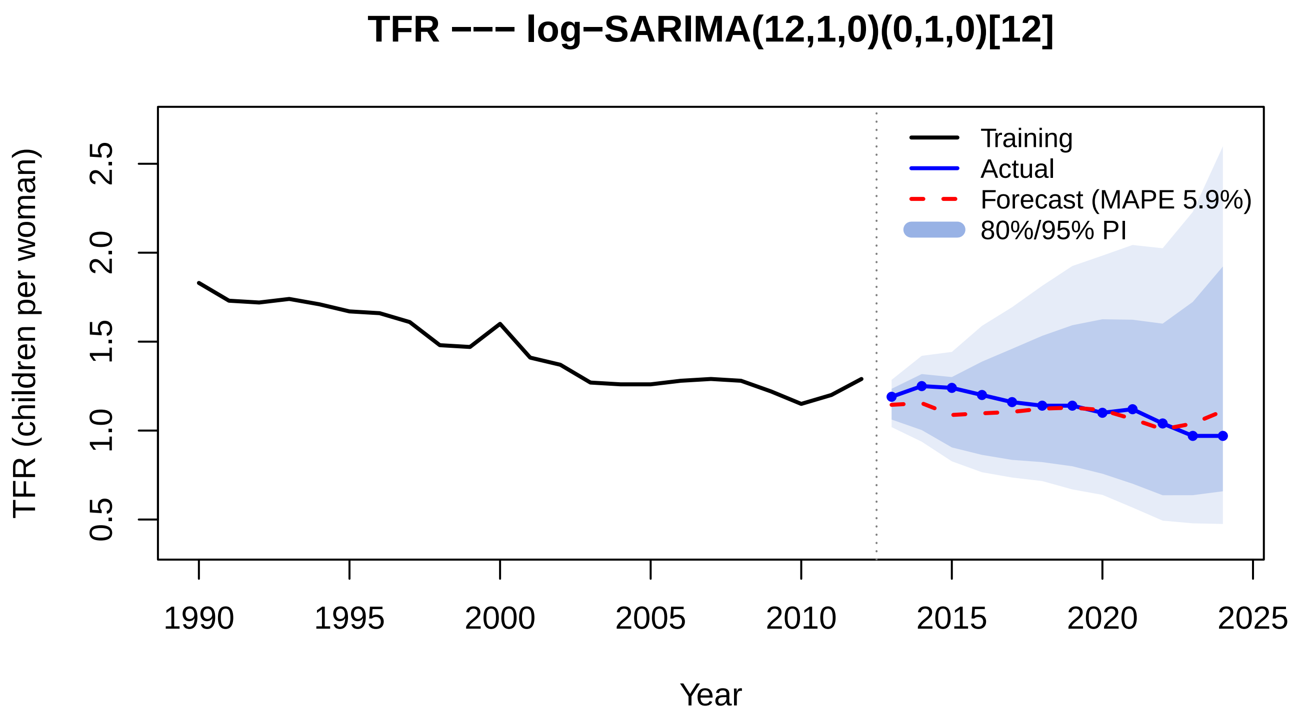


Figure 3: TFR: forecast vs actual, 2013–2024.

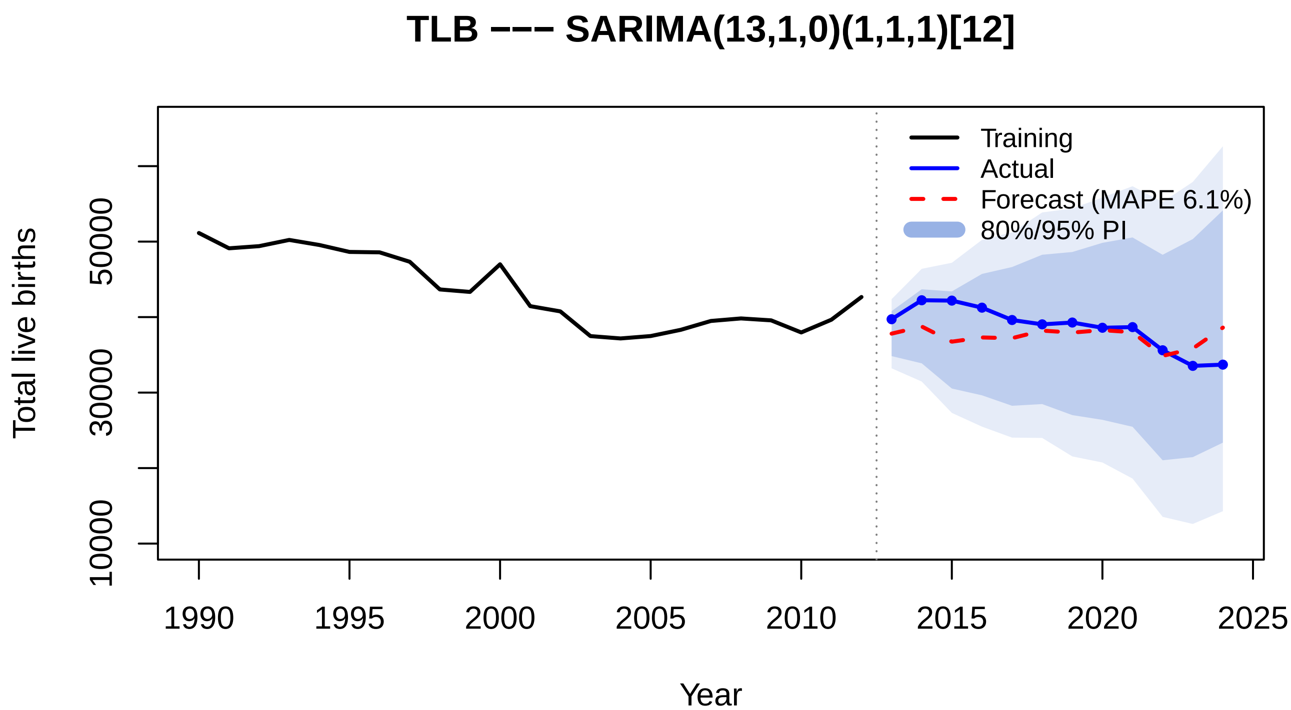


Figure 4: TLB: forecast vs actual, 2013–2024.